

Order Statistics of Exact First-Passage Target Coverage

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Abstract

We compute the exact first-passage order statistics of target coverage in a frozen sixteen-label, twenty-target subgroup model. Decoding all twenty C88 hit bitsets on all 65536 supports gives the exact target-coverage histogram at every prefix size. For each $r = 1, \dots, 20$, we determine the law and first four raw moments of the first time K_r at which at least r targets have been hit. Every distribution is carried by exact support counts and $16!$ -weighted permutation counts. The trivial target is retained, making $K_1 = 0$ deterministically. This is a finite result under `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Coverage statistic

Let $h_i(A)$ be the frozen C88 indicator that support A hits target H_i , and define

$$C(A) = \sum_{i=0}^{19} h_i(A), \quad K_r = \min\{k : C(A_k) \geq r\}.$$

Here A_k is the support of the first k labels in a uniform permutation. The trivial subgroup is target zero and satisfies $h_0(A) = 1$ for every A . Thus $C(\emptyset) = 1$ and $K_1 = 0$; all twenty targets remain in the model.

2 Exact laws

For $0 \leq k \leq 16$ and $1 \leq r \leq 20$, put

$$G_r(k) = \#\{A : |A| = k, C(A) \geq r\}.$$

Theorem 1. *The exact CDF and PMF of each coverage order statistic are*

$$\Pr(K_r \leq k) = \frac{G_r(k)}{\binom{16}{k}}, \quad \Pr(K_r = k) = \Pr(K_r \leq k) - \Pr(K_r \leq k-1).$$

Multiplying the reduced PMF by $16!$ gives a nonnegative integer permutation count. These counts sum to $16!$, and

$$\mathbb{E}[K_r] = \sum_{k=0}^{15} \Pr(K_r > k).$$

Moreover $K_r \leq K_{r+1}$ pointwise.

Proof. The first k positions of a uniform permutation have a uniform k -element support. Taking consecutive CDF differences gives the first-hit law. For $k \geq 1$, each oriented support edge that first crosses rank r has exactly $(k-1)!(16-k)!$ permutation completions; at time zero the empty support has $16!$ completions. The expectation is the standard finite survival sum, and nested threshold events give rank monotonicity. $\square$

3 Atlas features

The receipt partitions every one of the seventeen cardinality layers by all twenty possible coverage counts and records four raw moments, mean, and variance for every rank. For example, $\mathbb{E}[K_2] = 17/12$, while $\mathbb{E}[K_{10}] = 15589/3960$. The frozen support lattice admits coverage jumps: ranks 13 through 20 have the same law, equal to the top-target law, whose mean is $36499/3960$. These equalities are observations of the exact finite receipt, not a classification claim beyond it.

4 Audit and scope

An independent checker decodes every C88 bitset and reconstructs all support histograms. SymPy checks all twenty probability generating functions and their moments; clean replay preserves evidence hash

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Fifteen hostile mutations are rejected. No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside ring or table of marks, or Hilbert–Polya operator is claimed.